Assess the association between coffee intake and coffee types in mortality among individuals with abnormal P wave axis or negative P wave amplitude in lead V1.

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Abstract

Cardiovascular diseases are on the rise and diet has a significant impact on one's cardiovascular health. Coffee is an extremely popular beverage and an essential part of many people's diet around the globe, whose effects on health are not fully understood yet. Data from NHANES III was collected and studied to find a correlation between coffee intake with mortality in individuals with abnormal P wave axis or negative P wave amplitude. The type of coffee consumed was also considered, as the effects of caffeinated and decaffeinated coffee were thoroughly examined as well.

KEYWORDS:
coffee intake, caffeine consumption, mortality, P wave axis, negative P wave amplitude

1 | INTRODUCTION

Coffee is a staple of many people's diets and it has been playing an increasingly important role in their lives. Thus, it is no surprise that it has sparked interest in the scientific community and motivated it to study its effects on the human body. Moderate consumption of coffee has proved to have a positive impact on cognitive performance, memory and brain function, as well as enhance motor and respiratory functions. On the other hand, it may also be linked to nervousness and anxiety, among others[1]. Nonetheless, the effects of coffee on cardiovascular health remain unclear, as historically it was believed to have a negative impact, though recent studies mostly contradict those findings[2]. Globally, trends have shown that cardiovascular diseases have been on the rise over the last 30 years[3]. Abnormal P wave axis and abnormal negative P wave amplitudes are associated with an increased risk of arrhythmias and other cardiovascular complications, resulting in a higher cardiovascular mortality. Therefore, this study aims to investigate the association between coffee intake and coffee types in mortality, specifically among individuals with abnormal P wave axis or negative P wave amplitude in lead V1.

2 | METHODS

2.1 | Data Collection

The data used in this study were retrieved from the third National Health and Nutrition Examination Survey (NHANES III)[4], which collected data from 1988 to 1994, and from the National Center for Health Statistics (NCHS)[5]. This database was chosen since no other more recent NHANES had all the data required to conduct this investigation (i.e. coffee intake data, electrocardiogram data), or the corresponding data on mortality from the NCHS. Therefore, the findings from this study must be considered with caution, as the sample size is limited.

2.2 | Data Treatment

2.2.1 | Coffee Intake and Coffee Types

In this study, we have considered the following variables associated with coffee intake and coffee types. Coffee intake will be described by two main variables: COFFEE and CAFFEINE. The former describes the number of coffees an individual drinks in the past month, while the latter is the average quantity of caffeine consumed in a day in the past month, measured in milligrams (mg). It is important to note that these two variables may not necessarily be the same, since caffeine can also come from other food sources, such as green tea. Therefore, these two variables were considered separately in order to examine whether caffeine as a substance had more impact than the drink itself.

Regarding coffee types, all participants were divided into three groups, according to the type of coffee they consumed: caffeinated (TYPE = 0), decaffeinated (TYPE = 1), and both (TYPE = 2). Other ways of dividing the types of coffee were considered, such as sugar or sweetener, dairy or no dairy, but were ultimately discarded since the sample sizes were extremely limited and the results would not have any significant statistical significance.

2.2.2 | P Wave Axis and Negative P Wave Amplitude

Values for each participant's P wave axis (PAXIS) and negative P wave amplitude (PAMP1) were retrieved from the NHANES III database. The threshold used to define an abnormal P wave axis was < 0 degrees and > 75 degrees[6], whereas the threshold used to define an abnormal negative P wave amplitude was < -100 microvolts (μV)[7]. All individuals who met these criteria were selected and included in the investigation, meaning this study aims to investigate the effects of coffee solely on individuals who already present these abnormalities, either in their P wave axis or their negative P wave amplitude.

2.2.3 | Mortality

Regarding mortality, we have the following variables:

- Mortality Status: whether the individual is deceased as of 2019 (MORTSTAT)
  (i.e. 0 = Assumed alive, 1 = Assumed deceased)
- Cause of death (CAUSE)
  (i.e. 1 = Diseases of heart, 2 = Malignant neoplasms, 3 = Chronic lower respiratory diseases, 4 = Accidents, 5 = Cerebrovascular diseases, 6 = Alzheimer's disease, 7 = Diabetes mellitus, 8 = Influenza and pneumonia, 9 = Nephritis, nephrotic syndrome and nephrosis, 10 = All other causes, <NA> = Assumed alive, or no cause of death data available)
  For this study, only the individuals with CAUSE = 1 were considered, alongside the participants who are still alive, since the aim of the study is to investigate the effects of coffee on cardiovascular health
- Time (months) after the survey until the individual's death (TIME)
- Age (years) of the individual when they died (AGE.DEATH)

2.2.4 | Other Variables

Other variables were considered in order to better understand whether coffee had an actual impact on mortality, or if it was the result of other more prominent factors. Therefore, this study will also take into consideration the following factors:

- General information: Information about the individual's race, sex and age
- Smoker: Average number of cigarettes smoked in a day (CIGARETTES), and the time (years) of smoking this amount (TIME_SMOKING)
- Activity: Whether the individual considers themselves as more or less active than people their age (ACTIVE)
  (i.e. 1 = more active, 2 = less active, 3 = about the same)
- Hypertension: whether the individual was ever told by a doctor that they have hypertension (HYPERTENSION)
  (i.e. 1 = Yes, 2 = No)
- Diabetes: whether the individual was ever told by a doctor that they have diabetes (DIABETES)
  (i.e. 1 = Yes, 2 = No)

2.3 | Statistical Analysis

All statistical analyses were performed using R statistical software. Participants were categorized into different groups according to their consumption. Regarding coffee consumption (number of coffees consumed last month), the dataset was divided into 4 groups: None, Low (1-30), Moderate (30-99) and High (>99). On the other hand, participants were also divided according to their caffeine intake (average daily caffeine intake in the past month): None, Low (1-100mg), Moderate (101-300mg) and High (>300mg). Lastly, they were also grouped according to the type of coffee usually consumed: decaffeinated, caffeinated or both.

One-way ANOVA was used to compare participants' age at death across all consumption groups according to the categories, first we studied all groups divided by caffeine intake, then coffee consumption, and finally coffee types. In case there were significant differences between groups, Tukey's Honestly Significant Difference (HSD) test was used to identify which group pairs were significantly different. Also, a Kruskal–Wallis test was performed to verify the findings, since a non-parametric test may be more suited as the sample size is extremely small.

Afterwards, survival analysis was conducted using Kaplan-Meier survival curves, followed by log-rank tests to compare the curves. Then, the multivariate Cox proportional hazards model was used to quantify the differences while adjusting for cofounders (i.e. age, sex, hypertension, diabetes, smoking status, and physical activity level) and estimate 95% confidence intervals.

3 | RESULTS

3.1 | Age at death by caffeine and coffee consumption category on individuals with abnormal P wave axis

From the ANOVA, the age of death by caffeine consumption category was analized and the values obtained are shown in Table 1 . As we can observe, the p-value given by the analysis was 0.00621, meaning that there is a significant difference between groups.

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
caffeine category	3	1486	4195.2	4.189	0.00621
Residuals	368	43506	118.2		

TABLE 1 Anova table for age at death by caffeine consumption category

To further investigate this, we conducted a Tukey HSD test, and the results are shown in Table 2

Comparison	Difference	Lower CI	Upper CI	p-value
Low (1–100mg) – None	3.38	-1.39	8.14	0.261
Moderate (101–300mg) – None	1.79	-2.98	6.57	0.767
High (>300mg) – None	-2.01	-7.19	3.16	0.747
Moderate (101–300mg) – Low (1–100mg)	-1.58	-5.15	1.98	0.660
High (>300mg) – Low (1–100mg)	**-5.39**	**-9.47**	**-1.31**	**0.004**
High (>300mg) – Moderate (101–300mg)	-3.81	-7.90	0.29	0.079

TABLE 2 Tukey HSD test for age at death by caffeine consumption category

From the results shown above, we can conclude that the only statistically significant difference is between the low caffeine group and the high caffeine group. However, due to the limited sample size, further tests were conducted in order to better verify these results. Namely, a Kruskal–Wallis test was performed, as we believed that the small sample size could mean that a non-parametric test could better describe the relationship. The test yielded a p-value of 0.01163, confirming that there is a difference in groups. To find the pair of groups, we performed a pairwise Wilcoxon test, whose results are in Table 3 .

	None	Low (1–100mg)	Moderate (101–300mg)	High (>300mg)
Low (1–100mg)	0.6928	–		
Moderate (101–300mg)	1.0000	1.0000	–	
High (>300mg)	1.0000	**0.0089**	0.0965	–

TABLE 3 Pairwise Wilcoxon Test for Age of Death Across Caffeine Categories

The results from the pairwise Wilcoxon test agree with the findings of the Tukey HSD test, stating that the low caffeine group and the high caffeine group are statistically different.

However, caffeine categories in individuals with abnormal P wave axis are the only ones that show any difference between groups, as all other group categories show no difference between themselves. Figure 1 shows us a boxplot graph with data about the four different groups of coffee consumption in individuals with abnormal P wave axis. We can observe that, intuitively, it appears the different groups do not have a significant difference in the age of death, and the data support this intuition.

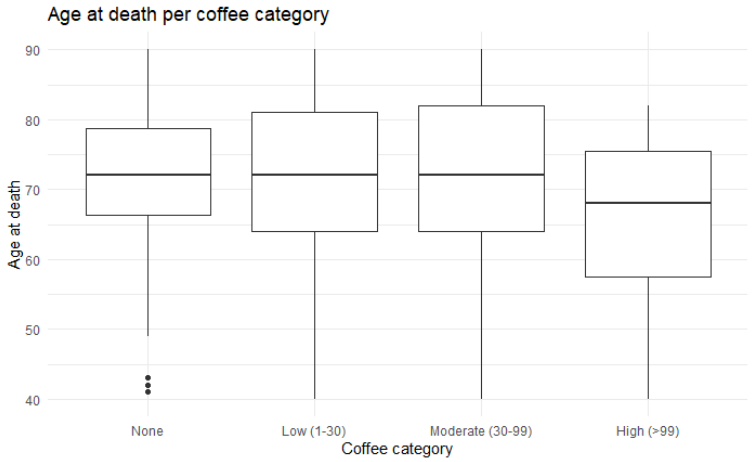


FIGURE 1 Individuals with abnormal P wave axis.

For example, Table 4 shows that the p-value from the age of death by coffee consumption categories is 0.204, which is not statistically significant.

Source	**Df**	**Sum Sq**	**Mean Sq**	**F value**	**Pr(>F)**
coffee category	3	558	185.9	1.54	0.204
Residuals	368	44434	120.7		

TABLE 4 Anova table for age at death by coffee consumption category (PAXIS)

All other group categories follow the same trend (their respective ANOVA tables and boxplots can be found in Appendix 7.1.), leading us to believe that the amount of caffeine, coffee or the type of coffee consumed may not have, overall, a significant impact on cardiovascular health.

3.2 | Survival curves by category

In Figure 2 we can see a Kaplan-Meier survival curve displaying unadjusted survival probabilities across the four categories of average daily caffeine intake over a follow-up period of up to 400 months.

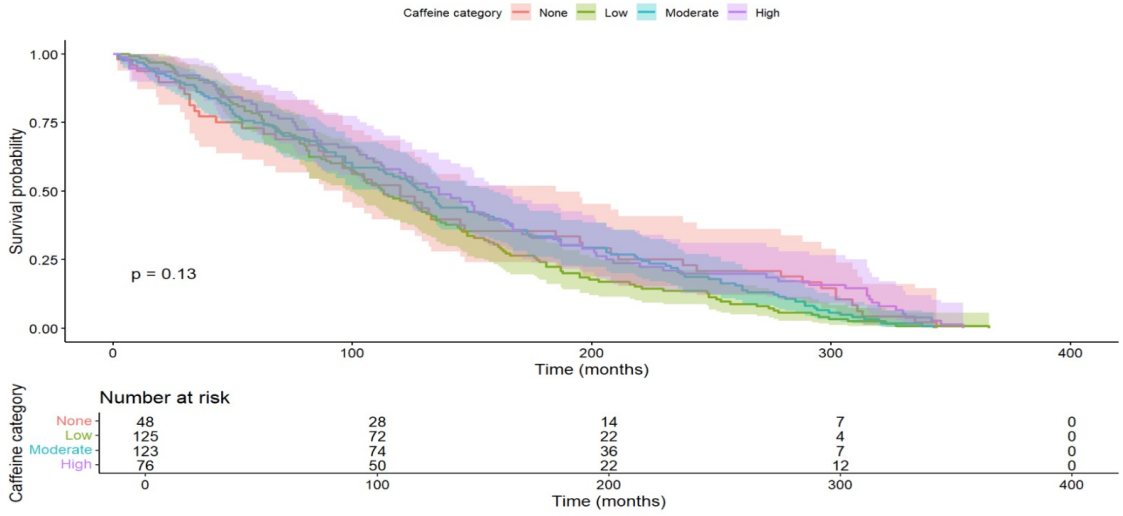


FIGURE 2 Kaplan-Meier survival curves of caffeine intake groups for individuals with abnormal P wave axis.

All four curves follow a similar downward trajectory, with no consistent or pronounced separation. Confidence bands for all groups overlap substantially, indicating no substantial differences. The log-rank test supports this, giving a p-value of 0.1, meaning it has no significant statistical value. Overall, these results suggest no evidence of a survival benefit or detriment associated with different levels of coffee consumption in individuals with abnormal P wave axis.

Figure 3 presents adjusted survival curves for each group of average daily caffeine intake derived from a multivariable Cox proportional hazards model.

The analysis takes into consideration the following cofounders: smoking status, activity level, age, sex, diabetes and hypertension, and survival probabilities are standardized for a representative patient profile (i.e., average age, male, non-smoker, inactive, without diabetes or hypertension). We can observe that all four curves are tightly clustered with overlapping confidence intervals, which suggests no meaningful difference in adjusted survival across groups. This supports the conclusion that, after taking all other variables into consideration, caffeine consumption does not independently influence mortality risk in individuals with abnormal P wave axis.

Therefore, we can conclude that both unadjusted and adjusted survival analyses revealed no evidence of a significant association between caffeine consumption and mortality among individuals with abnormal P wave axis. The unadjusted Kaplan-Meier curves showed minimal visual separation across the four intake groups and the log-rank test confirmed this with a non-significant result. In the adjusted Cox model, after controlling for all other variables, caffeine intake was not independently associated with survival. However, it showed strong and statistically significant association between other covariates and mortality. For instance, heavy smoking had a hazard ratio (HR) of 2.61 with a p-value of 0.0001, diabetes had an HR of 1.52 and a p-value of 0.007 and age had an HR of 1.09 with a p-value $< 2 \times 10^{-16}$. Thus, we can confidently conclude that caffeine intake does not influence mortality risk in this population, whereas clinical and behavioral factors play a more prominent role in survival outcomes.

Regarding all other categories, all the survival curves follow the same trend as described above, there are no significant diversions from the analysis made on the survival curves for caffeine intake, and, therefore, there is no need for an in-depth discussion for all of them. All the graphs are available in the Appendix 7.2. There is, however, one aspect that should be pointed out. For example, in Figure 12 , we can observe that the confidence band for category "Both" may appear to slightly diverge from the cluster of confidence bands, yet this can be justified by the excessively small simply size, $n = 11$ at TIME $= 0$ and $n = 1$ at TIME $= 300$. All other occurrences of this phenomenon can be traced back to the same justification.

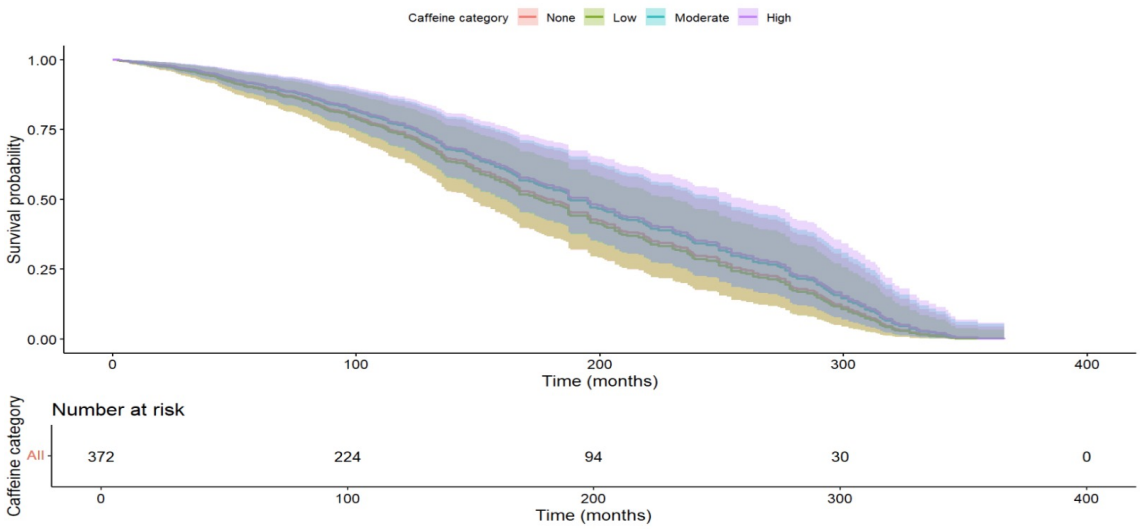


FIGURE 3 Multivariable Cox proportional hazards model survival curves of caffeine intake groups for individuals with abnormal P wave axis.

4 | DISCUSSION

Initial results from the ANOVA and Tukey HSD tests suggested a significant difference in age at death across caffeine consumption groups among individuals with abnormal P wave axis. More specifically, the only statistically significant difference emerged between the Low (1-100 mg/day) and High (>300 mg/day) caffeine intake groups. This finding was reinforced by the non-parametric Kruskal–Wallis test and pairwise Wilcoxon test, which also identified this pairwise difference as significant. However, no other pairwise comparisons were significant, and all ANOVA analyses on coffee quantity and coffee type yielded non-significant p-values, suggesting no meaningful impact of these variables on age at death.

Despite these early signals, survival analysis provided a more comprehensive and nuanced perspective. Both the Kaplan–Meier survival curves and the log-rank test showed no significant differences in survival probabilities across caffeine intake groups, with curves tightly overlapping and a log-rank p-value of 0.1. This lack of significant divergence was consistent in all other group analyses (i.e. coffee quantity and coffee type), and was also observed in individuals with negative P wave amplitude, as detailed in the appendix.

The results from the multivariable Cox proportional hazards model, which controlled for major confounding factors such as age, sex, diabetes, hypertension, smoking status, and physical activity, were particularly informative. In this adjusted model, caffeine intake was not independently associated with survival, and confidence intervals for all caffeine groups overlapped substantially. Instead, classical cardiovascular risk factors such as age, diabetes, and especially heavy smoking showed strong and statistically significant associations with mortality. For instance, smoking had a hazard ratio (HR) of 2.61 ($p < 0.001$), diabetes had an HR of 1.52 ($p = 0.007$), and age had a particularly strong effect (HR = 1.09; $p < 2 \times 10^{-16}$).

Taken together, these findings suggest that while there may be minor unadjusted differences in age of death across caffeine intake groups, these differences do not translate into survival benefit or harm once relevant covariates are controlled for. This supports the broader literature, which generally indicates that coffee consumption, in particular in moderate quantities, has neutral to beneficial effects on cardiovascular health[8,9], and is unlikely to be harmful in patients with existing cardiovascular risk factors[10,11].

However, all findings from this study should be taken with caution. An important limitation of this study is the limited sample size, especially for certain subgroups such as participants who consumed both caffeinated and decaffeinated coffee. In those cases, confidence intervals were wider and survival curves showed greater fluctuation, which may be due to random variation rather than any true effect. Furthermore, coffee intake was self-reported, and the measurement of caffeine exposure may not fully reflect long-term habits or other dietary confounders (e.g., sugar, milk, or other additives).

Ultimately, the study does not support a causal relationship between coffee intake and mortality in individuals with abnormal P wave axis or negative P wave amplitude. Rather, it highlights that traditional risk factors such as smoking, diabetes, and age remain the strongest predictors of survival outcomes in this population. Future studies with a larger sample size and more controlled methods of data collection may help clarify whether any subgroups of coffee consumers benefit or are harmed by their intake.

5 | CONCLUSION

This study explored the potential association between coffee intake and cogge type, and mortality in individuals with abnormal P wave parameters, which reflects a population at elevated cardiovascular risk. While initial analyses indicated a minor difference in age at death between low and high caffeine consumers, these findings did not hold under survival analysis or multivariable adjustment.

The comprehensive Cox regression models confirmed that classical risk factors such as age, diabetes, and smoking status remain the dominant predictors of mortality, with no independent association observed for coffee or caffeine intake. Furthermore, Kaplan–Meier curves and log-rank tests supported the absence of survival differences across coffee consumption groups.

In light of these results, coffee consumption does not appear to significantly impact mortality in this high-risk subgroup. These findings reinforce the notion that public health interventions should continue to focus on tackling other more prominent issues such as smoking and diabetes, rather than restricting coffee consumption.

6 | BIBLIOGRAPHY

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7 | APPENDICES

7.1 | Age at death by category

7.1.1 | Boxplots

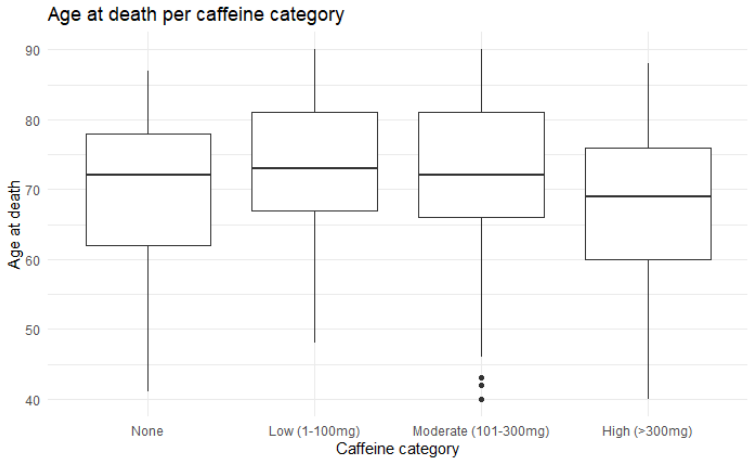


FIGURE 4 Individuals with abnormal P wave axis.

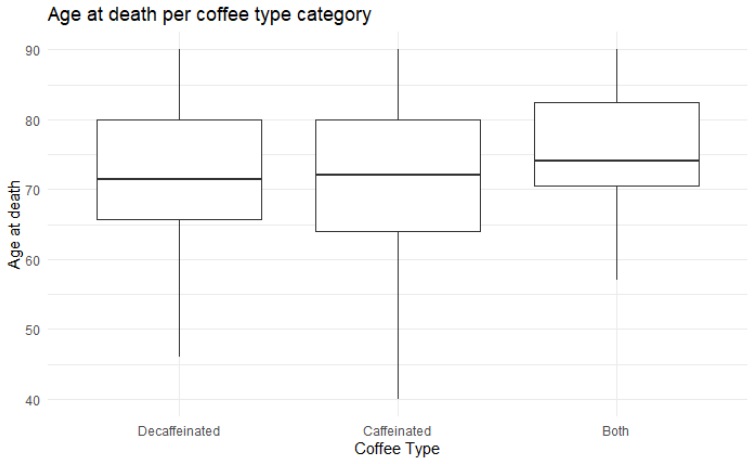


FIGURE 5 Individuals with abnormal P wave axis.

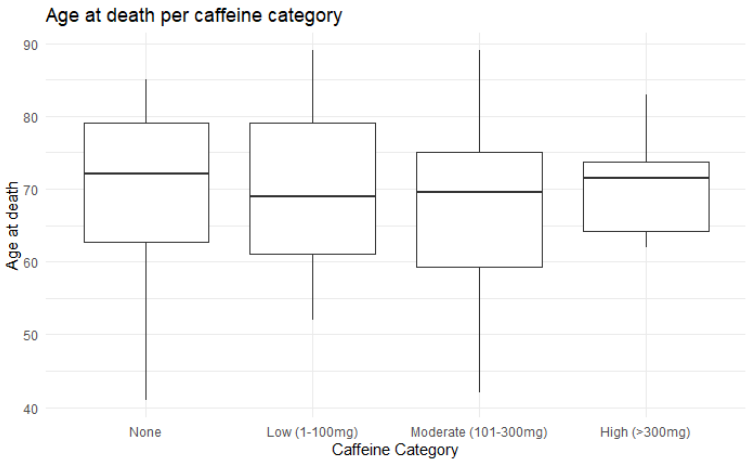


FIGURE 6 Individuals with abnormal negative P wave amplitude.

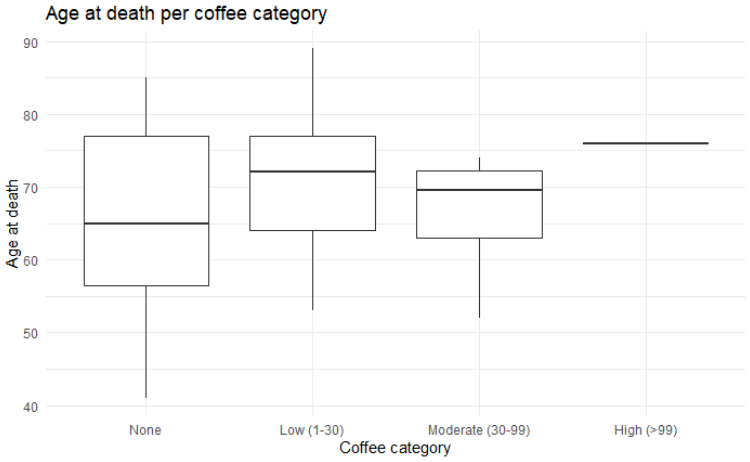


FIGURE 7 Individuals with abnormal negative P wave amplitude.

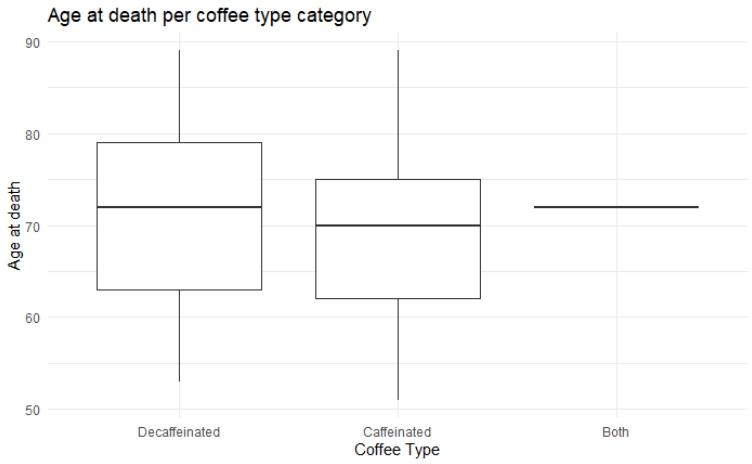


FIGURE 8 Individuals with abnormal negative P wave amplitude.

7.1.2 | ANOVA tables

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
coffee types category	2	299	149.3	1.203	0.302
Residuals	269	33383	124.1		

TABLE 5 Anova table for age at death by coffee types consumption category (PAXIS)

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
caffeine category	3	109	36.21	0.311	0.818
Residuals	69	8043	116.57		

TABLE 6 Anova table for age at death by caffeine consumption category (PAMP)

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
coffee category	3	548	182.6	1.657	0.184
Residuals	69	7604	110.2		

TABLE 7 Anova table for age at death by coffee consumption category (PAMP)

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
coffee types category	2	38	19.15	0.191	0.827
Residuals	47	4724	100.51		

TABLE 8 Anova table for age at death by coffee types consumption category (PAMP)

7.2 | Survival Curves

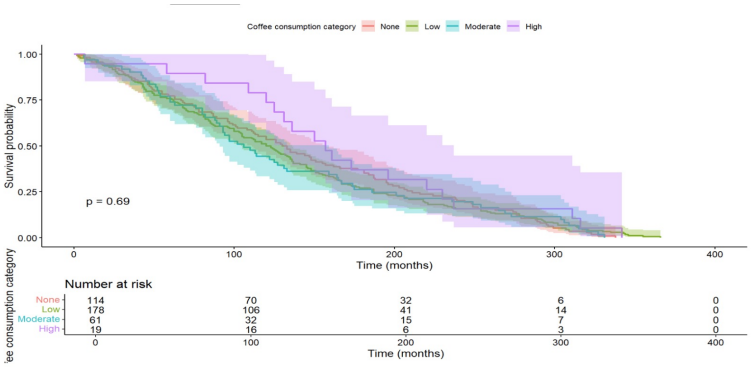


FIGURE 9 Kaplan-Meier survival curves of coffee consumption groups for individuals with abnormal P wave axis.

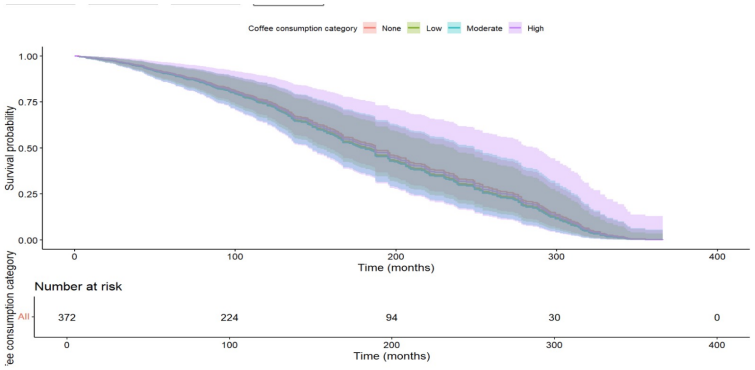


FIGURE 10 Multivariable Cox proportional hazards model survival curves of coffee consumption groups for individuals with abnormal P wave axis.

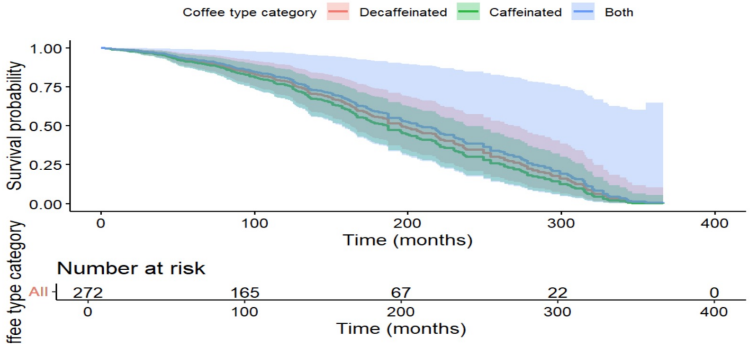


FIGURE 11 Multivariable Cox proportional hazards model survival curves of coffee types groups for individuals with abnormal P wave axis.

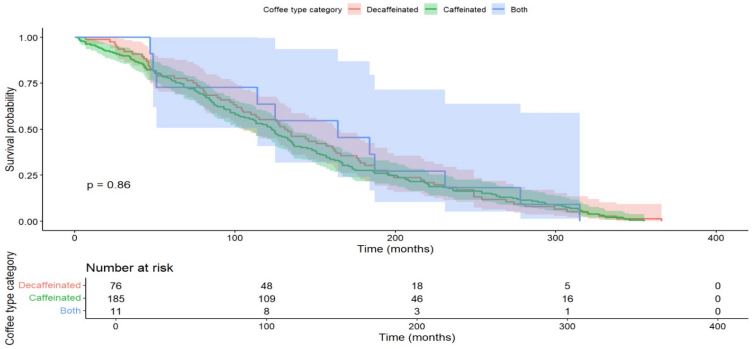


FIGURE 12 Kaplan-Meier survival curves of coffee types groups for individuals with abnormal P wave axis.

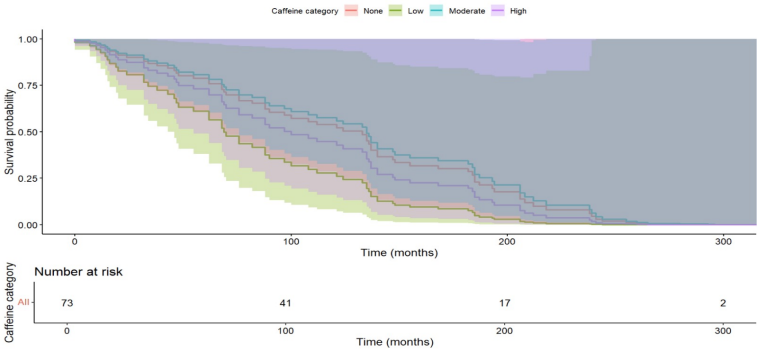


FIGURE 13 Multivariable Cox proportional hazards model survival curves of caffeine intake groups for individuals with abnormal negative P wave amplitude.

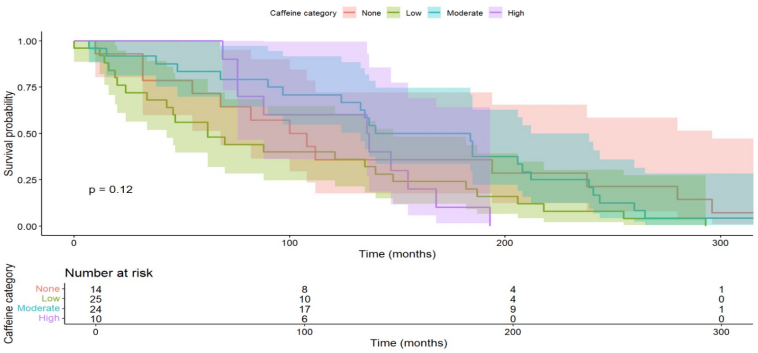


FIGURE 14 Kaplan-Meier survival curves of caffeine intake groups for individuals with abnormal negative P wave amplitude.

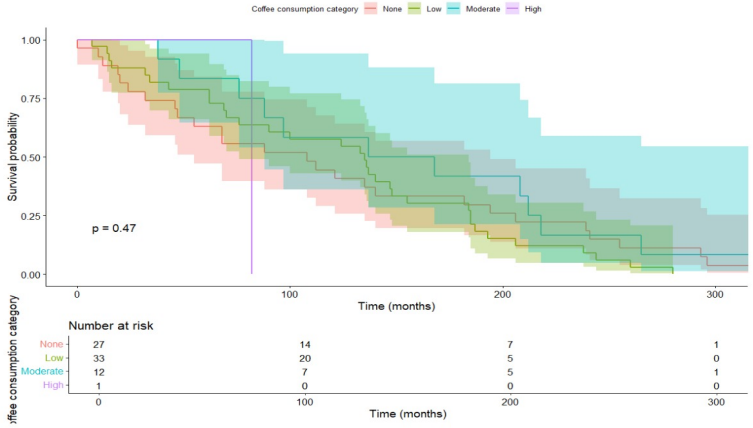


FIGURE 15 Kaplan-Meier survival curves of coffee consumption groups for individuals with abnormal negative P wave amplitude.

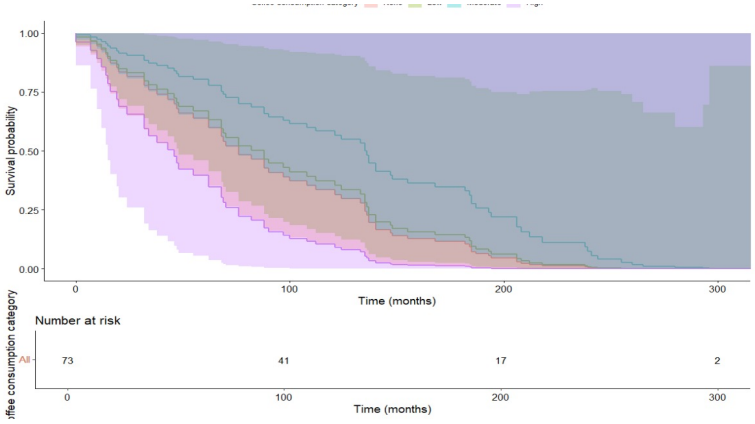


FIGURE 16 Multivariable Cox proportional hazards model survival curves of coffee consumption groups for individuals with abnormal negative P wave amplitude.

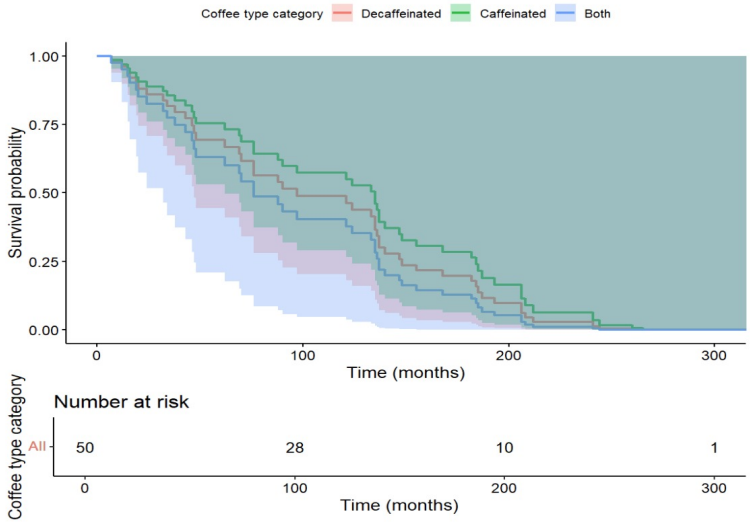


FIGURE 17 Multivariable Cox proportional hazards model survival curves of coffee types groups for individuals with abnormal negative P wave amplitude.

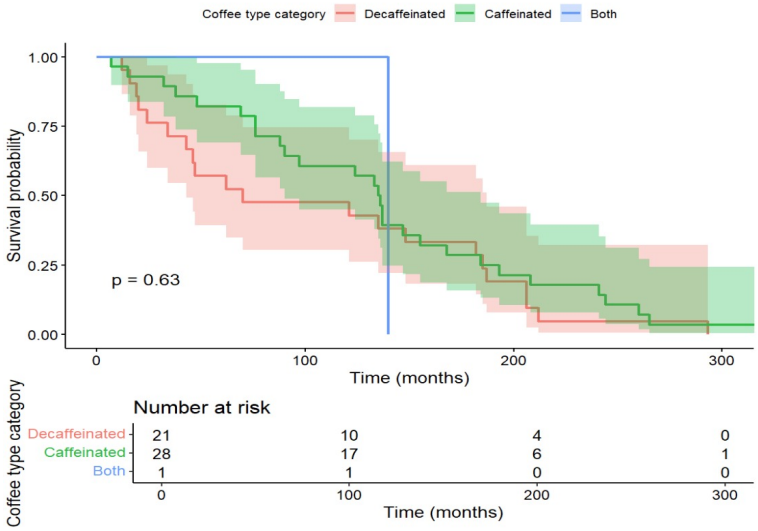


FIGURE 18 Kaplan-Meier survival curves of coffee types groups for individuals with abnormal negative P wave amplitude.